STEVE: ROS 2 Autonomous Robot Butler & Mobile Manipulator

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Core Developers: Shrikar Nakhye · Pratik Adhikari · Riddhesh More | **Workspace:** `steve_ros2_ws` (ROS 2 Humble)

ROS 2 Humble | Neobotix MMO-700 | UR5e 6-DoF | Nav2 + A* | RTAB-Map 3D | SAM3 + DeepSeek

Robot Platform & Hardware Specs

Mobile Base: Neobotix MMO-700 Omnidirectional (4-wheel omnidrive)

Manipulator: Universal Robots UR5e 6-DoF Arm + Robotiq 2F-85 Gripper

Perception Head: Intel RealSense L515 on Dynamixel XL430-W250 Pan-Tilt Unit

Safety LiDAR: Dual SICK S300 Laser Scanners (360° planar safety coverage)

System Bringup: `ROS_AUTOSTART.sh` in `screen` daemon; static IP / DHCP

Layered Software Architecture

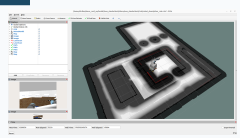


- `steve_essentials`: Hardware drivers (Base, LiDAR, L515), `neo_msgs2`, kinematics.
- `steve_hardware_bringup`: Synchronized startup of base, arm, and sensors without race conditions.
- `steve_simulation`: High-fidelity Gazebo digital twin with URDF/Xacro parity.
- **Docker AI Services:** Isolated GPU containers for heavy AI models (Mask3D, SAM3).

Simulation & Visualization Environments



Gazebo (`steve_house`)



Nav2 + SLAM in RViz2

- **Simulation Worlds:** `steve_house`, `small_house`, and `neo_workshop`.
- **Deployment Profiles:** Config profiles for simulation, hardware, and rosbag replay.

Autonomy, Perception & Reasoning Stack

1. Autonomous Navigation & Localization (Nav2)

- **A* Global Planning:** Replaced default Dijkstra with A* heuristic for rapid, optimal path search in narrow domestic and workshop passages.
- **AMCL Particle Filter Tuning:** Increased particle thresholds and update rates to eliminate robot kidnapping in symmetric, feature-sparse environments.
- **Sim-to-Real Transfer:** Unified TF tree and topics for seamless sim-to-real code reuse.

2. 3D Semantic Mapping & Scene Graphs (RTAB-Map)

- **Bandwidth Optimization:** Compressed `RGBDImage` transport (JPEG+Uint16) achieving ≈90% serialization reduction (0.8 MB vs 8 MB/frame).
- **RANSAC + DBSCAN:** Horizontal plane detection ($\mathbf{n} \cdot [0, 0, 1] \geq 0.95$) with 2D plane-tangent clustering to cleanly isolate floors from obstacles.
- **Scene Graph Generation:** PCA-based Oriented Bounding Boxes (OBB), support planes ($\mathbf{n}_z > 0.9$), and containment relations for domestic objects.

3. Open-Vocabulary Command Grounding (LLM + SAM3)

- **DeepSeek-R1 Commonsense LLM:** Infers likely target locations (e.g., "water bottle → kitchen table") when objects are unindexed in scene graph.
- **SAM3 Zero-Shot Vision:** Containerized prompt-based visual segmentation and 3D raycasting to project camera detections into global map coordinates.
- **Collision-Aware Standoff:** FOV-based standoff position optimization with Open3D point-cloud collision checks for unobstructed surface inspection.

Key Engineering Achievements & Benchmarks

- **Network Bandwidth:** 90% DDS serialization reduction via compressed RGBD feeds.
- **TF Sanitization Relay:** Fixed historical bag jumps/orphans, raising replay from 40% to **95%**.
- **GPU Acceleration:** Docker GPU passthrough slashed 3D segmentation from **40m** to **2m**.
- **PTU SyncWrite Driver:** Simultaneous Dynamixel packet writes eliminate inter-motor jitter.
- **Hands-Off Operation:** Full autostart daemon boots all drivers safely on physical hardware.

Team Contributions & Work Package Breakdown

Shrikar Nakhye

- Core architecture & modular migration
- Centralized bringup & auto-start daemon
- Nav2 A* global planner & AMCL tuning
- Hardware repair, network & Git dev workflow

Pratik Adhikari

- RTAB-Map 3D RGB-D SLAM pipeline
- RANSAC + DBSCAN plane segmentation
- Scene graph generation & spatial relations
- TF sanitization relay (95% replay)

Riddhesh More

- DeepSeek-R1 + SAM3 grounding pipeline
- Dynamixel XL430 SyncWrite controller
- Collision-aware standoff visual planner
- Nav2 Action Wrapper & Gazebo worlds